

Discrete Math — Monday Test

Worked Solutions (Rosen, 7th ed.)

Question numbers match the companion Practice Problems PDF exactly.

Tuesday — §1.1 & §1.3

1. Negations:

- (a) Mei does *not* have an MP3 player.
- (b) There *is* pollution in New Jersey.
- (c) $2 + 1 \neq 3$.
- (d) The summer in Maine is not hot and sunny (i.e., it is not hot, or it is not sunny — De Morgan's law applied to English).

2.

- (a) $p \wedge q$
- (b) $p \wedge \neg q$
- (c) $\neg p \wedge \neg q$
- (d) $p \vee q$
- (e) $p \rightarrow q$
- (f) $(p \vee q) \wedge (p \rightarrow \neg q)$
- (g) $p \leftrightarrow q$

3.

- (a) $T \rightarrow F = \mathbf{False}$
- (b) $F \rightarrow T = \mathbf{True}$
- (c) $F \rightarrow F = \mathbf{True}$
- (d) $F \rightarrow F = \mathbf{True}$

4. These are the **Identity**, **Domination**, and **Idempotent** laws. Each holds because a truth table check over both values of p shows the two sides always match:

(a),(b): identity element leaves p unchanged. (c),(d): F/T dominate $\wedge/\vee$ respectively. (e),(f): combining p with itself changes nothing.

5. $\neg(a \vee b) \equiv \neg a \wedge \neg b$ and $\neg(a \wedge b) \equiv \neg a \vee \neg b$:

- (a) Kwame will not take a job in industry and will not go to graduate school.
- (b) Yoshiko does not know Java or does not know calculus.
- (c) James is not young or not strong.
- (d) Rita will not move to Oregon and will not move to Washington.

6. **Not** a tautology. Build the truth table:

p	q	$p \rightarrow q$	$\neg p$	$\neg p \wedge (p \rightarrow q)$	result
T	T	T	F	F	T
T	F	F	F	F	T
F	T	T	T	T	F
F	F	T	T	T	T

When $p = F, q = T$: the premise $\neg p \wedge (p \rightarrow q)$ is true but $\neg q$ is false, so the conditional is false.
Counterexample: $p = F, q = T$.

7. Using the key identity $p \rightarrow q \equiv \neg p \vee q$:

$$\neg q \rightarrow \neg p \equiv \neg(\neg q) \vee \neg p \equiv q \vee \neg p \equiv \neg p \vee q \equiv p \rightarrow q. \quad \blacksquare$$

8.

- (a) **Satisfiable.** Try $p = F, q = F$: $(F \vee T) \wedge (T \vee F) \wedge (T \vee T) = T \wedge T \wedge T = T$.
- (b) **Unsatisfiable.** $p \rightarrow q$ and $p \rightarrow \neg q$ can't both be true unless $p = F$. Then $\neg p \rightarrow q$ forces $q = T$, but $\neg p \rightarrow \neg q$ forces $q = F$ — contradiction.
- (c) **Unsatisfiable.** $p \leftrightarrow q$ requires p, q equal; $\neg p \leftrightarrow q$ requires p, q opposite. Can't both hold.

Wednesday — §1.4

9.

- (a) $P(0)$: $0 \leq 4$ — **True**
- (b) $P(4)$: $4 \leq 4$ — **True**
- (c) $P(6)$: $6 \leq 4$ — **False**

10.

- (a) $\exists x (P(x) \wedge Q(x))$
- (b) $\exists x (P(x) \wedge \neg Q(x))$
- (c) $\forall x (P(x) \vee Q(x))$
- (d) $\forall x (\neg P(x) \wedge \neg Q(x))$ (equivalently $\neg \exists x (P(x) \vee Q(x))$)

11. This is the domain-trap exercise — compare each part carefully.

- (a) $\exists x (C(x) \wedge D(x) \wedge F(x))$ — note "a student" means *some* student, so this is $\exists$, not $\forall$.
- (b) $\forall x (C(x) \vee D(x) \vee F(x))$ — "all students" $\Rightarrow \forall$.
- (c) $\exists x (C(x) \wedge F(x) \wedge \neg D(x))$
- (d) $\neg \exists x (C(x) \wedge D(x) \wedge F(x))$ (equivalently $\forall x \neg (C(x) \wedge D(x) \wedge F(x))$)
- (e) $\exists x C(x) \wedge \exists x D(x) \wedge \exists x F(x)$

12.

- (a) True on: the set of students currently enrolled in a discrete math course. False on: the general population.
- (b) True on: residents of a retirement home (all over 21). False on: a kindergarten class.
- (c) True on: a set of full biological siblings (they share one mother). False on: a random group of unrelated classmates.
- (d) True on: a set of unrelated people from different families. False on: a set of first cousins on the maternal side (they can share a grandmother).

13. Let the domain be drivers / movies / people / students as appropriate.

- (a) Original: $\exists x \neg P(x)$. Negation: $\forall x P(x)$ — "All drivers obey the speed limit."
- (b) Original: $\forall x (S(x) \rightarrow P(x))$. Negation: $\exists x (S(x) \wedge \neg P(x))$ — "There is a Swedish movie that is not serious."
- (c) Original: $\forall x \neg P(x)$. Negation: $\exists x P(x)$ — "Someone can keep a secret."
- (d) Original: $\exists x \neg P(x)$. Negation: $\forall x P(x)$ — "Everyone in this class has a good attitude."

Thursday — §1.5 & start §1.6

14. Watch how the English meaning changes with quantifier order.

- (a) Some student has sent an email to some student (possibly themselves).
- (b) There is a student who has sent an email to every student.
- (c) Every student has sent an email to at least one student.
- (d) There is a student who has received an email from every student.
- (e) Every student has received an email from at least one student.
- (f) Every student has sent an email to every student.

15.

- (a) $\forall x L(x, \text{Jerry})$
- (b) $\forall x \exists y L(x, y)$
- (c) $\exists y \forall x L(x, y)$
- (d) $\neg \exists x \forall y L(x, y)$
- (e) $\exists y \neg L(\text{Lydia}, y)$
- (f) $\exists y \forall x \neg L(x, y)$
- (g) $\exists y (\forall x L(x, y) \wedge \forall z [(\forall x L(x, z)) \rightarrow z = y])$
- (h) $\exists x \exists y (x \neq y \wedge L(\text{Lynn}, x) \wedge L(\text{Lynn}, y) \wedge \forall z (L(\text{Lynn}, z) \rightarrow (z = x \vee z = y)))$
- (i) $\forall x L(x, x)$
- (j) $\exists x \forall y ((y \neq x) \rightarrow \neg L(x, y))$

16.

- (a) There is a number (namely 0) that, added to any number, leaves that number unchanged — i.e. an additive identity exists.
- (b) For any nonnegative x and negative y , $x - y$ is positive.
- (c) There exist nonpositive numbers x and y whose difference is positive (e.g. $x = 0, y = -1$).
- (d) The product of two real numbers is zero if and only if at least one of the numbers is zero.

17. Push negations inward using De Morgan's laws for quantifiers ($\neg \forall \equiv \exists \neg$, $\neg \exists \equiv \forall \neg$):

- (a) $\forall y \forall x \neg P(x, y)$
- (b) $\exists x \forall y \neg P(x, y)$
- (c) $\forall y (\neg Q(y) \vee \exists x R(x, y))$
- (d) $\forall y (\forall x \neg R(x, y) \wedge \exists x \neg S(x, y))$
- (e) $\forall y (\exists x \forall z \neg T(x, y, z) \wedge \forall x \exists z \neg U(x, y, z))$

18. Argument form: $p \rightarrow q, p \therefore q$. This is **Modus Ponens** — a valid form. Yes: if both premises are true, the conclusion must be true.

19.

- (a) **Addition**
- (b) **Simplification**
- (c) **Modus Ponens**
- (d) **Modus Tollens**
- (e) **Hypothetical Syllogism**

Friday — §1.6 (finish)

20.

- (a) **Simplification**
- (b) **Disjunctive Syllogism**
- (c) **Modus Ponens**
- (d) **Addition**
- (e) **Hypothetical Syllogism**

21. Let p : play hockey, q : sore, r : use whirlpool, etc.

- (a) $p \rightarrow q$, $q \rightarrow r$, $\neg r$. Modus Tollens on $q \rightarrow r$, $\neg r$ gives $\neg q$; Modus Tollens again on $p \rightarrow q$, $\neg q$ gives **“I did not play hockey.”**
- (b) The premises talk about different days (Monday/Friday worked, Tuesday not sunny, Friday not partly sunny) that don’t chain together. **No definite conclusion follows** — this item is testing whether you notice the premises don’t actually connect.
- (c) From “all insects have six legs” + “dragonflies are insects” (Hypothetical Syllogism via universal instantiation): **“All dragonflies have six legs.”** From “spiders do not have six legs” + the universal (contrapositive): **“Spiders are not insects.”**
- (d) From $\forall x(\text{Student}(x) \rightarrow \text{Internet}(x))$ and $\neg \text{Internet}(\text{Homer})$, Modus Tollens gives **“Homer is not a student.”** (Maggie’s fact is extra, unused information.)
- (e) From “healthy $\rightarrow$ doesn’t taste good” + “tofu is healthy” (Modus Ponens): tofu doesn’t taste good. Combined with “you only eat what tastes good” (contrapositive: doesn’t taste good $\rightarrow$ don’t eat it), Modus Tollens gives **“you do not eat tofu”** — already stated as a premise, so no new conclusion follows about cheeseburgers (the healthy $\rightarrow$ bad-taste rule says nothing about non-healthy foods).
- (f) Disjunctive Syllogism on “dreaming or hallucinating” + “not dreaming” gives “hallucinating.” Modus Ponens then gives **“I see elephants running down the road.”**

22.

- (a) $\text{JAVA}(\text{Doug})$ [premise]. $\forall x(\text{JAVA}(x) \rightarrow \text{HighPay}(x))$ [premise]. **Universal Instantiation** at Doug, then **Modus Ponens** gives $\text{HighPay}(\text{Doug})$. **Existential Generalization** gives the conclusion.
- (b) $\exists x(\text{Class}(x) \wedge \text{Whale}(x))$ [premise]. **Existential Instantiation** to a specific c . **Simplification** gives $\text{Whale}(c)$. **Universal Instantiation** + **Modus Ponens** on the second premise gives $\text{CaresOcean}(c)$. **Simplification** again gives $\text{Class}(c)$; **Conjunction** combines them; **Existential Generalization** gives the conclusion.
- (c) **Universal Instantiation** on both universal premises at Zeke, then two applications of **Modus Ponens** (own PC $\rightarrow$ use word processor) give the conclusion.
- (d) **Existential Instantiation** on the second premise to a specific c ; **Simplification** gives $\text{NJ}(c)$ and $\neg \text{SeenOcean}(c)$; **Universal Instantiation** + **Modus Ponens** on the first premise gives $\text{Within50}(c)$; **Conjunction**, then **Existential Generalization** gives the conclusion.

Saturday — §1.7 & §1.8

23. Direct proof. Let m, n be odd: $m = 2a + 1$, $n = 2b + 1$ for integers a, b . Then

$$m + n = 2a + 2b + 2 = 2(a + b + 1),$$

which is $2 \times$ an integer, hence even. ■

24. Proof by contradiction. Let x be irrational and y rational. Suppose, for contradiction, $x + y$ is rational, say $x + y = r$. Then $x = r - y$. Since r, y are both rational and the rationals are closed under subtraction, x would be rational — contradicting that x is irrational. So $x + y$ must be irrational. ■

25. Proof by contraposition. Contrapositive: if m and n are both odd, then mn is odd. Let $m = 2a + 1$, $n = 2b + 1$:

$$mn = (2a + 1)(2b + 1) = 4ab + 2a + 2b + 1 = 2(2ab + a + b) + 1,$$

which is odd. Since the contrapositive holds, the original statement holds. ■

26. $P(1)$: $(a + b)^1 \geq a^1 + b^1$, i.e. $a + b \geq a + b$ — true by equality. **This is a trivial proof:** $P(1)$ reduces immediately to an identity, so it's proven directly without needing induction machinery.

27. Counterexample: $n = 7$. The only squares ≤ 7 are 0, 1, 4. Checking every combination of three: $4 + 1 + 1 = 6$, $4 + 4 + 1 = 9$, $4 + 1 + 0 = 5$, $1 + 1 + 1 = 3$, $4 + 0 + 0 = 4$, $1 + 1 + 0 = 2$, $1 + 0 + 0 = 1$, $0 + 0 + 0 = 0$ — none equal 7. So 7 cannot be written as the sum of three squares.

28. Proof by exhaustion (only 4 cases to check):

n	$n^2 + 1$	$2n$	holds?
1	2	2	✓
2	5	4	✓
3	10	6	✓
4	17	8	✓

All four cases hold, so the statement is proven for $1 \leq n \leq 4$. ■

29. Proof by cases on the ordering of a, b, c . E.g. Case $a \leq b \leq c$: $\min(b, c) = b$, so LHS = $\min(a, b) = a$. Also $\min(a, b) = a$, so RHS = $\min(a, c) = a$ (since $a \leq c$). Both sides equal a . The same check across all six orderings shows both expressions always equal $\min(a, b, c)$, so they're equal in every case. ■

30. Constructive. Take $n = 1$: the sum of the positive integers not exceeding 1 is just 1, and indeed $1 = 1$. We exhibited the number explicitly, so the proof is constructive.

31. Nonconstructive. $2 \cdot 10^{500} + 15$ and $2 \cdot 10^{500} + 16$ are consecutive integers. Consecutive integers greater than 1 can never both be perfect squares (consecutive squares grow further apart than 1 once past 0, 1). So at least one of the two must fail to be a perfect square — but we never determine *which* one.

32. Proof by contradiction, mirroring the $\sqrt{2}$ proof. Suppose $\sqrt[3]{2} = p/q$ in lowest terms ($\gcd(p, q) = 1$). Then $2q^3 = p^3$, so p^3 is even, so p is even (odd³ is odd). Write $p = 2k$: $8k^3 = 2q^3 \Rightarrow q^3 = 4k^3$, so q^3 is even, so q is even. But then p, q share the factor 2, contradicting $\gcd(p, q) = 1$. Hence $\sqrt[3]{2}$ is irrational. ■

Sunday — §5.1 Mathematical Induction

33.

(a) $P(1)$: $1^2 = \frac{1(1+1)(2 \cdot 1 + 1)}{6}$.

(b) $\text{LHS} = 1$; $\text{RHS} = \frac{1 \cdot 2 \cdot 3}{6} = 1$. Equal, so $P(1)$ is true.

(c) Inductive hypothesis: assume $1^2 + 2^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}$ for some positive integer k .

(d) Need to prove $P(k+1)$: $1^2 + \cdots + k^2 + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}$.

(e)

$$\begin{aligned} \frac{k(k+1)(2k+1)}{6} + (k+1)^2 &= (k+1) \left[\frac{k(2k+1)}{6} + (k+1) \right] \\ &= (k+1) \cdot \frac{2k^2 + 7k + 6}{6} = \frac{(k+1)(k+2)(2k+3)}{6}. \quad \checkmark \end{aligned}$$

(f) Since $P(1)$ holds and $P(k) \Rightarrow P(k+1)$ for every positive integer k , by the principle of mathematical induction $P(n)$ holds for all positive integers n . ■

34. **Base case** $n = 1$: $\text{LHS} = 1$, $\text{RHS} = (-1)^0 \cdot \frac{1 \cdot 2}{2} = 1$. ✓

Inductive hypothesis: assume true for $n = k$. **Inductive step:**

$$\text{LHS}_{k+1} = (-1)^{k-1} \frac{k(k+1)}{2} + (-1)^k (k+1)^2 = (-1)^k (k+1) \left[(k+1) - \frac{k}{2} \right] = (-1)^k \frac{(k+1)(k+2)}{2},$$

which matches the formula at $n = k+1$. By induction, true for all positive integers n . ■

35.

(a) $P(2)$: $2! < 2^2$.

(b) $2! = 2 < 4 = 2^2$. True, so $P(2)$ holds.

(c) Inductive hypothesis: assume $k! < k^k$ for some integer $k > 1$.

(d) Need to prove $(k+1)! < (k+1)^{k+1}$.

(e) $(k+1)! = (k+1) \cdot k! < (k+1) \cdot k^k < (k+1) \cdot (k+1)^k = (k+1)^{k+1}$, using the IH and $k^k < (k+1)^k$ since $k < k+1$.

(f) Since $P(2)$ holds and $P(k) \Rightarrow P(k+1)$, by induction $n! < n^n$ for all integers $n > 1$. ■

36. **Base case** $n = 5$: $2^5 = 32 > 25 = 5^2$. ✓

Inductive step: assume $2^k > k^2$ for some $k \geq 5$. Then $2^{k+1} = 2 \cdot 2^k > 2k^2$. For $k \geq 5$: $2k^2 - (k+1)^2 = k^2 - 2k - 1 \geq 25 - 10 - 1 = 14 > 0$, so $2k^2 > (k+1)^2$. Chaining: $2^{k+1} > 2k^2 > (k+1)^2$. By induction, true for all integers $n > 4$. ■

37. **Base case** $n = 1$: $1^2 + 1 = 2$, divisible by 2. ✓

Inductive hypothesis: $k^2 + k = 2m$ for some integer m . **Inductive step:**

$$(k+1)^2 + (k+1) = (k^2 + k) + (2k + 2) = 2m + 2k + 2 = 2(m + k + 1),$$

divisible by 2. By induction, true for all positive integers n . ■

(Quick alternate proof: $n^2 + n = n(n + 1)$, the product of two consecutive integers — one of them is always even.)

38. This is the single most important "spot the error" pattern for Monday. The flaw is in the inductive step's claim that the two groups of k horses "overlap." This is only true when $k \geq 2$ (each group then has at least one horse in common). But the very first application of the inductive step goes from $P(1)$ to $P(2)$, where $k = 1$: the "first $k = 1$ horses" is just horse 1, and the "last $k = 1$ horses" is just horse 2 — two *disjoint* single-horse sets that share **no** horse. So the overlap argument silently fails exactly at the base transition, and the chain of implications never actually gets started. The inductive step is not valid for all $k \geq 1$ — only for $k \geq 2$ — which breaks the whole induction.